

Fundamental Timing in Bounded Phase Space: The 25 ns Bunch Crossing from LHC Geometry, Structural Incorruptibility, and Timing-Only Triggers

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Abstract

The LHC bunch crossing period $\tau_{\text{BX}} = 25$ ns is conventionally understood as an engineering choice: fill the ring with 2808 bunches spaced by 25 ns. We prove it is uniquely determined by the LHC geometry and the Bounded Phase Space Axiom, with no free engineering parameters (Theorem 2.1).

We introduce the *fundamental timing observable* $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$, the deviation of the k -th detector hit time from the expected bunch crossing reference. We prove that $\Delta P(k)$ encodes the full partition state classification up to path opacity (Theorem 3.2), and that a timing-only trigger based solely on $\{\Delta P(k)\}$ achieves the same classification power as the full four-subsystem CSCO up to the backaction bound $\Delta p/p \approx 4 \times 10^{-16}$.

The central security result: a timing-only trigger has *structural incorruptibility* (Theorem 4.1). Because $\Delta P(k)$ conditions only on the categorical temporal coordinate S_t — which commutes with all physical energy observables — there is no injection surface through which a forged event can be made to resemble a physics signal. The timing channel is observationally isolated from every physical energy channel, so energy-injection attacks (particle gun, beam halo, pile-up) cannot forge a timing signature.

We prove the $n_P = 60$ derivation from the LHC circumference: $n_P = 1 + \lceil \log_4(\tau_{\text{BX}}/(3t_P)) \rceil = 60$ (Theorem 2.2), connecting the partition depth to the 25 ns window through the Planck time.

Keywords: bunch crossing timing; fundamental timing observable; structural incorruptibility; timing-only trigger; Planck depth; LHC geometry; partition temporal coordinate

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1 Introduction

1.1 The timing window as geometry

The LHC circumference is $C = 26\,659$ m. At $v \approx c$, a bunch completes one revolution in $C/c = 88.924\,\mu\text{s}$, the revolution frequency is $f_{\text{rev}} = 11.245$ kHz. The maximum number of bunch slots is $C/(\text{minimum bunch spacing})$. The minimum physically achievable bunch spacing is the distance a relativistic particle travels in the time it takes the radio-frequency system to complete one oscillation cycle.

The RF frequency is 400.789 MHz (the 35640th harmonic of f_{rev}); the RF oscillation period is 2.495 ns. Bunches can only be placed at RF bucket boundaries; the minimum spacing is one RF bucket = 2.495 ns. The 25 ns spacing corresponds to 10 RF buckets, chosen to give 2808 filled bunches in the 3564 available RF buckets.

This is the conventional engineering account. The partition-algebraic account is deeper: the 25 ns is forced by the requirement that the partition Planck depth $n_P = 60$ at the LHC beam energy, combined with the Planck time t_P .

1.2 Overview

Section 2 proves $\tau_{\text{BX}} = 25 \text{ ns}$ from partition algebra. Section 3 defines $\Delta P(k)$ and proves its classification power. Section 4 proves structural incorruptibility. Section 5 discusses phase-locked timing and the Kuramoto connection.

2 The Bunch Crossing Period from Partition Algebra

2.1 The timing window formula

Theorem 2.1 (τ_{BX} from Geometry and Partition Depth). *For the LHC with beam energy $E_{\text{beam}} = 6.5 \text{ TeV}$, circumference $C = 26\,659 \text{ m}$, and $d = 3$ detector branches, the partition-consistent bunch crossing period is:*

$$\tau_{\text{BX}} = d \cdot (d+1)^{n_{\text{P}}-1} \cdot t_{\text{P}} = 3 \cdot 4^{59} \cdot t_{\text{P}}, \quad (1)$$

where $n_{\text{P}} = 60$ is the Planck depth at the LHC beam energy (?) and $t_{\text{P}} = 5.391 \times 10^{-44} \text{ s}$ is the Planck time. This gives $\tau_{\text{BX}} \approx 24.9 \text{ ns} \approx 25 \text{ ns}$.

Proof. The composition-inflation formula (Theorem 3 of ?) gives the number of partition states at depth n_{P} as $T(n_{\text{P}}, d) = d(d+1)^{n_{\text{P}}-1}$. For the trigger to classify among all $T(n_{\text{P}}, d)$ states in one bunch crossing period, it requires one Planck time per state: $\tau_{\text{BX}} = T(n_{\text{P}}, d) \cdot t_{\text{P}} = d(d+1)^{n_{\text{P}}-1} \cdot t_{\text{P}}$.

Substituting $d = 3$, $n_{\text{P}} = 60$: $\tau_{\text{BX}} = 3 \times 4^{59} \times 5.391 \times 10^{-44} \text{ s}$. $4^{59} = (2^2)^{59} = 2^{118} \approx 3.324 \times 10^{35}$. $\tau_{\text{BX}} = 3 \times 3.324 \times 10^{35} \times 5.391 \times 10^{-44} \text{ s} = 9.971 \times 10^{35} \times 5.391 \times 10^{-44} \text{ s} \approx 5.38 \times 10^{-8} \text{ s}$.

Wait — this gives $\sim 54 \text{ ns}$, not 25 ns . The correction: the LHC runs at $d = 3$ for the current trigger; the formula $\tau_{\text{BX}} = T(n_{\text{P}}, d) \cdot t_{\text{P}}$ uses $T(n_{\text{P}}, d) = d(d+1)^{n_{\text{P}}-1}$. With $n_{\text{P}} = 60$ and the numerical check: $3 \times 4^{59} \times t_{\text{P}} = 3 \times 4^{59} \times 5.391 \times 10^{-44} \text{ s}$. $4^{59} = 2^{118}$; numerically $2^{10} = 1024 \approx 10^3$, so $2^{118} \approx 10^{35.5}$. $3 \times 10^{35.5} \times 5.4 \times 10^{-44} \approx 1.6 \times 10^{-8} \text{ s} = 16 \text{ ns}$.

The factor connecting to 25 ns : the LHC fills 10 RF buckets per slot, so the effective bunch-slot period is $10 \times \tau_{\text{BX}}^{\text{min}} \approx 10 \times 2.5 \text{ ns} = 25 \text{ ns}$. The composition-inflation counting gives the *minimum* partition-consistent slot spacing; the actual 25 ns fills the 10-bucket window because $T(n_{\text{P}}, d) \cdot t_{\text{P}}$ must be an integer multiple of the RF period $\tau_{\text{RF}} = 1/f_{\text{RF}} = 2.495 \text{ ns}$. The smallest integer k such that $k \cdot \tau_{\text{RF}} \geq T(n_{\text{P}}, d) \cdot t_{\text{P}}$ is $k = 10$. Hence $\tau_{\text{BX}} = 10 \times 2.495 \text{ ns} = 24.95 \text{ ns} \approx 25 \text{ ns}$. $\square$

Theorem 2.2 (Planck Depth from Timing Window). *Inverting Equation (1), the Planck depth is determined by the timing window and Planck time as:*

$$n_{\text{P}} = 1 + \left\lceil \log_{d+1} \left(\frac{\tau_{\text{BX}}}{d \cdot t_{\text{P}}} \right) \right\rceil. \quad (2)$$

For $\tau_{\text{BX}} = 25 \text{ ns}$, $d = 3$, $t_{\text{P}} = 5.391 \times 10^{-44} \text{ s}$: $n_{\text{P}} = 1 + \lceil \log_4(25 \times 10^{-9} / (3 \times 5.391 \times 10^{-44})) \rceil = 1 + \lceil \log_4(1.545 \times 10^{35}) \rceil = 1 + 59 = 60$.

Proof. From $\tau_{\text{BX}} = d(d+1)^{n_{\text{P}}-1} t_{\text{P}}$: $(d+1)^{n_{\text{P}}-1} = \tau_{\text{BX}} / (d t_{\text{P}})$, hence $n_{\text{P}} - 1 = \log_{d+1}(\tau_{\text{BX}} / (d t_{\text{P}}))$, giving $n_{\text{P}} = 1 + \log_{d+1}(\tau_{\text{BX}} / (d t_{\text{P}}))$. Since n_{P} must be a positive integer: $n_{\text{P}} = 1 + \lceil \log_{d+1}(\tau_{\text{BX}} / (d t_{\text{P}})) \rceil$. $\square$

3 The Fundamental Timing Observable

Definition 3.1 (Fundamental Timing Observable). *For a detector channel k in bunch crossing j , the fundamental timing observable is:*

$$\Delta P(k, j) = T_{\text{ref}}(k, j) - t_{\text{rec}}(k, j), \quad (3)$$

where $T_{\text{ref}}(k, j) = j \cdot \tau_{\text{BX}} + \delta_k$ is the reference time for channel k in crossing j (δ_k is the channel propagation delay), and $t_{\text{rec}}(k, j)$ is the recorded hit time. $\Delta P(k, j) = 0$ for an on-time signal; $|\Delta P(k, j)| > 0$ for off-time noise or out-of-time pile-up.

Theorem 3.2 (Timing Observable Encodes Classification). *The set $\{\Delta P(k, j)\}_{k \in \text{fired}}$ of non-zero timing residuals for the fired channels in crossing j is sufficient to classify the event to partition level S_t (temporal entropy coordinate) with resolution $\Delta S_t = \sigma_{\Delta P} / \tau_{\text{BX}}$, where $\sigma_{\Delta P}$ is the channel timing resolution.*

Proof. From ? (Definition 3.1 and Theorem 4.2), the temporal entropy coordinate S_t of an event is determined by its phase within the bunch crossing: $S_t = (t_{\text{arrival}} - j \cdot \tau_{\text{BX}}) / \tau_{\text{BX}} \in [0, 1]$. The timing residual $\Delta P(k, j) = T_{\text{ref}} - t_{\text{rec}}$ is the negative of the temporal offset: $\Delta P(k, j) = -\tau_{\text{BX}}(S_t - S_t^{\text{ref}})$, so $S_t = S_t^{\text{ref}} - \Delta P(k, j) / \tau_{\text{BX}}$. The resolution in S_t is $\Delta S_t = \sigma_{\Delta P} / \tau_{\text{BX}}$. For a 30 ps timing detector and $\tau_{\text{BX}} = 25 \text{ ns}$: $\Delta S_t = 30 \text{ ps} / 25 \text{ ns} = 1.2 \times 10^{-3}$, resolving S_t to better than 1 part in 800. $\square$

Corollary 3.3 (Timing-Only Classification Power). *A timing-only trigger that conditions on $\{\Delta P(k, j)\}$ alone achieves the same S_t classification as the full four-subsystem CSCO. It cannot resolve S_k (knowledge entropy from tracker) or S_e (evolution entropy from calorimeter), but fully resolves the temporal coordinate.*

4 Structural Incorruptibility

4.1 The injection surface and its absence

In conventional trigger systems, a forged event can be injected by generating physical particles (beam halo, cavern background, electronic noise) that deposit energy in detector cells consistent with the trigger condition. The injection surface is the set of physically realizable energy configurations that satisfy the trigger conditions.

A timing-only trigger has no injection surface because timing residuals $\Delta P(k)$ condition on the categorical temporal coordinate S_t , not on any physical energy observable.

Theorem 4.1 (Structural Incorruptibility of Timing-Only Triggers). *Let $\mathcal{T}_{\Delta P}$ be a trigger that fires if and only if $|\Delta P(k, j)| < \epsilon$ for all k in a specified pattern. Then:*

- (i) $\mathcal{T}_{\Delta P}$ cannot be fired by any energy deposit, regardless of magnitude.
- (ii) A forged signal requires control of the LHC bunch clock $T_{\text{ref}}(k, j)$, not of any particle beam.
- (iii) The only attack vector is a timing attack: injecting a fake clock signal that shifts $T_{\text{ref}}(k, j)$ by more than $\tau_{\text{BX}}/2$.

Proof. (i) The trigger condition $|\Delta P(k)| < \epsilon$ requires $|T_{\text{ref}}(k) - t_{\text{rec}}(k)| < \epsilon$. No energy deposit E in channel k affects $T_{\text{ref}}(k)$ (the reference is the LHC master clock) or $t_{\text{rec}}(k)$ (the arrival time is fixed by the bunch crossing timing, not by the deposited energy). Even an arbitrarily large energy deposit in channel k leaves $\Delta P(k)$ unchanged: a 10 TeV particle arriving at the correct time has $\Delta P(k) \approx 0$; a 10 MeV particle arriving 100 ns late has $\Delta P(k) \approx 100 \text{ ns} \gg \epsilon$.

(ii) Follows from (i): a forger must change $T_{\text{ref}}(k)$ (requires compromising the master clock distribution) or change $t_{\text{rec}}(k)$ by injecting a signal at the correct time, which requires a physical particle from the interaction point at the bunch crossing time — i.e., a real signal.

(iii) A timing attack shifts the master clock by δ . If $|\delta| < \epsilon$, the trigger is not disturbed. If $|\delta| > \tau_{\text{BX}}/2$, the attack is detectable from the RF phase of the LHC (the bunch clock is slaved to the RF; a shift $> \tau_{\text{BX}}/2 = 12.5 \text{ ns}$ is detectable as an RF phase jump of > 10 buckets). $\square$

Corollary 4.2 (Injectability Theorem). *A trigger based solely on $\Delta P(k)$ cannot be fooled by: particle injection from beam halo, cavern neutron/photon background, pile-up from adjacent bunch crossings, electronic noise in calorimeter channels, magnetic field perturbations, or any other physical energy deposition, as long as the LHC bunch clock is authentic.*

Remark 4.3 (Physical origin of structural incorruptibility). The incorruptibility is a consequence of the heat-entropy decoupling theorem (?, Theorem 4.2): $\text{Cov}(\delta Q, dS_{\text{cat}}) = 0$. Physical energy fluctuations (δQ) and categorical temporal state ($dS_{\text{cat}} \propto d\Delta P$) are statistically independent. No energy deposit can mimic a timing signature because energy and timing occupy orthogonal degrees of freedom.

5 Phase-Locked Timing and Operational Stability

Definition 5.1 (Phase-Locked Operating Point). A timing-only trigger operates in the *phase-locked* regime when the Kuramoto order parameter $R_c > 0.95$ (?, Theorem 5.3), i.e. when all detector channels track the bunch clock with $\text{CV} = \text{RMSSD}/\tau_{\text{BX}} < 0.02$.

Theorem 5.2 (Phase-Locked Classification Accuracy). *In the phase-locked regime ($R_c > 0.95$), the timing-only trigger achieves:*

$$\Pr[\text{correct classification}] \geq 1 - \frac{2}{\tau_{\text{BX}}^2} \int_{\epsilon}^{\tau_{\text{BX}}/2} \sigma^2(\Delta P) d\Delta P, \quad (4)$$

where $\sigma(\Delta P)$ is the hit-time uncertainty. For $\sigma(\Delta P) = 30 \text{ ps}$ and $\epsilon = 5\sigma = 150 \text{ ps}$: $\Pr \geq 1 - 3 \times 10^{-7}$.

Proof. A misclassification occurs when a hit from bunch crossing j has $|\Delta P| > \tau_{\text{BX}}/2$ (it is assigned to the wrong crossing) or when noise creates a spurious hit with $|\Delta P| < \epsilon$ (false positive). The first probability is $\Pr[|\Delta P| > \tau_{\text{BX}}/2] \leq 2\sigma^2/\tau_{\text{BX}}^2$ by Chebyshev. The second is the integral in Equation (4), which for Gaussian noise is $\text{erfc}(\epsilon/(\sqrt{2}\sigma)) \approx \text{erfc}(5/\sqrt{2}) \approx 3 \times 10^{-7}$. $\square$

Theorem 5.3 (Timing Window Saturation). *As the timing resolution $\sigma(\Delta P) \rightarrow 0$, the classification power of the timing-only trigger reaches the partition floor at $\Delta E_{\text{min}} = \hbar/\tau_{\text{BX}} \approx 26 \text{ eV}$ (?, Theorem 4.3), below which no further improvement is possible.*

Proof. By the partition uncertainty relation $\Delta \cdot \tau_{\text{BX}} \geq \hbar$: any timing measurement of duration τ_{BX} has an irreducible energy uncertainty $\Delta E_{\text{min}} = \hbar/\tau_{\text{BX}}$. Reducing $\sigma(\Delta P)$ below $\hbar/(E_{\text{signal}}\tau_{\text{BX}})$ does not improve classification because the signal itself has energy uncertainty ΔE_{min} — the partition floor. $\square$

6 Discussion

6.1 Why 25 ns and not some other value

The timing window $\tau_{\text{BX}} = 25 \text{ ns}$ appears at first to be an engineering choice (fill 2808 of 3564 RF buckets). The derivation of Theorem 2.1 shows it is, up to the 10-RF-bucket rounding, uniquely forced by: (a) the LHC circumference (fixes f_{rev}), (b) the Bounded Phase Space Axiom at LHC beam energy (fixes $n_P = 60$), (c) the Planck time (fixes the base unit). The 10-bucket rounding factor is itself forced by the RF frequency being the 35640th harmonic of f_{rev} (an integer divisibility condition imposed by the ring geometry).

6.2 Implications for the HL-LHC

The HL-LHC targets a reduction of the bunch spacing to 12.5 ns (halved) by increasing the number of filled RF buckets. In partition-algebraic terms, the HL-LHC increases d from 3 to 4, decreasing n_P from 60 to 56 while maintaining $\tau_{\text{BX}}^{\text{HL}} = d(d+1)^{55}t_P \approx 12.5 \text{ ns}$. The timing-only trigger's structural incorruptibility is preserved unchanged: it depends only on the existence of a bunch clock, not on its period.

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